Joel E. Castro Hernandez

joel.e.castro@usc.edu | (619) 610-8132 | joel-ca.github.io/portfolio

Education

University of Southern California, Viterbi School of Engineering Aug 2026 – May 2028
M.S. Computer Science

University of California, Berkeley Jul 2022 – May 2026
B.A. Computer Science, SEED Scholar Honors Program, GPA: 3.5

Research Experience

Stanford Robotics and Embodied AI Lab (REAL) [Prof. Shuran Song] Mar 2025 – Aug 2025

- Co-authored a CoRL conference publication with Prof. Shuran Song and Stanford graduate students [1]
- Trained and deployed Diffusion Policies as baselines against our novel policy architecture [1]
- Designed a novel robotic handwriting task, including a 3D-printed pen gripper, for handheld data collection (collecting 450+ demonstrations), transferred to ARX5 robot for rollout [1]
- Built an evaluation pipeline leveraging Grounded-SAM2 and OpenCV for benchmarking robot handwriting [1]

UCSF Cardiac Vision Laboratory [Prof. Jan Christoph] Sep 2024 – Jan 2025

- Worked with SoTA differential rendering 3D image processing methods to optimize visualization techniques for cardiac optical and ultrasound data
- Developed a multi-view camera calibration and visualization tool, automating the calibration process, eliminating repeated manual execution, and minimizing downtime by 90%

CMU Software and Societal Systems Dept. [Prof. Jonathan Aldrich] May 2024 – Aug 2024

- First-authored SPLASH publication on a type theory for enhancing diagramming tools [2]
- Categorically coded 150+ open-source visualization tools' source code (e.g., AMD GPUOpen, Torchview) for their data structure decomposition properties, yielding statistically significant evidence of our proposed type theory [2]

Berkeley Staller Labs [Prof. Max Staller] Aug 2022 – May 2024

- Explored SHAP analysis on an in-lab convolutional neural network to increase its interpretability by determining which features it weighs most when predicting activation domains in proteins [3]
- Streamlined all-atom Monte Carlo simulation pipeline of disordered proteins on a computer cluster with Bash scripting, leading to a time-saving increase of over 85%, and wrote documentation used in training 2 peers

Publications

1. A. Patel, B. Pekarek, **J. E. C. Hernandez**, and S. Song, "Behavior prompting policy: Demonstrations as prompts for manipulation," *arXiv preprint arXiv:2606.30457*, 2026 [Webpage] [Paper]
2. **J. E. Castro Hernandez** and O. G. John, "Understanding program visualizations in the wild," in *Companion Proceedings of the 2024 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity (SPLASH Companion '24)*, pp. 49–51, 2024 [Webpage] [Paper]
3. C. LeBlanc, P. Agarwal, J. Demaray, G. Hu, M. Zintel, A. Lam, **J. E. Castro Hernandez**, and M. Staller, "Interpretable biophysical neural networks of transcriptional activation domains separate roles of protein abundance and coactivator binding," *bioRxiv*, 2025 [Paper]

Projects

Minecraft Fabric Mod – CS184 Final Project Honorable Mention | **Website** Jan 2025 – May 2025

- Implemented procedural sphere generation, texture mapping, and quaternion-based rotation mechanics for physically accurate interactions
- Contributed to shader enhancements, video production, and final deliverables, helping the project earn Honorable Mention in CS184

Sockrates: Clothing Sorting/Folding (on 7-DOF Robot Arm) | **Website** Aug 2024 – Dec 2024

- Led the development of Python/OpenCV categorization scripts, resulting in 100% classification accuracy
- Built ROS Publisher/Subscriber System: integrated CV and precise robotic actuation, implementing forward and inverse kinematics

Teaching Experience

Computer Graphics (CS184) TA [1 semester] - UC Berkeley — Dec 2025 – May 2026

- Performed weekly office hours and graded technical homework and exams on topics such as ray tracing, physical simulation, and more

Undergraduate Graphics Group (UCBUGG) Facilitator [3 semesters] - UC Berkeley — Jan 2025 – May 2026

- Presented weekly lectures and mentored student groups through end-to-end production of animated shorts

Computer Architecture (CS61C) Tutor/Grader [1 semester] - UC Berkeley — Aug 2024 – Dec 2024

- Addressed an average of 30+ student questions weekly during office hours on C, RISC-V, and Logisim concepts

CSM: CS70 Junior Mentor [1 semester] - UC Berkeley — Jan 2024 – May 2024

- Instructed a 1-unit supplementary course, providing support in discrete math & probability

Presentations

Stanford SURF Symposium 2025: Lightning talk and poster presentation of Behavior Prompting Policy.

SPLASH 2024 (Pasadena, CA): Poster and oral presentation of *Understanding Program Visualizations in the Wild.*

SACNAS 2024 (Phoenix, AZ): Poster presentation of *Understanding Program Visualizations in the Wild.*

Academic Service

(Berkeley) Anova – On-site Tutor; Publicity/Curriculum Committee — Jan 2023 – May 2026

- Taught coding concepts in Python and Scratch to high school students weekly, enhancing their understanding of CS
- Developed Arduino curricula to improve tech accessibility for students in under-resourced communities

College of Computing, Data Science, and Society (CDSS) Advisory Board — Aug 2023 – May 2024

- 1 of 9 undergraduates invited to work with CDSS deans to develop, disseminate, and analyze surveys to seek advice from a 100-person Undergraduate Student Advisory Board

Awards

- **Jack Kent Cooke Graduate Scholarship -** Highly competitive national scholarship for graduate studies — 2026
- **CODE2040 Fellowship -** Fellowship for top Black and Latinx computer science students — 2024
- **eBay Pathways Fellowship -** Fellowship supporting underrepresented students in tech — 2024
- **Jack Kent Cooke Scholarship -** Highly competitive national scholarship for undergraduate studies — 2022
- **Cal Alumni Leadership Award -** Merit-based scholarship recognizing leadership at UC Berkeley — 2022
- **SEED Scholars Honors Program -** STEM honors program for academic excellence and research — 2022

Skills

Languages: Python, C/C++, Java, Bash, SQL, JavaScript/HTML/CSS

Software & Tools: ROS, Unreal Engine, Git, Arduino, Logisim

Design: Autodesk (Maya, Fusion, Inventor), Adobe (Photoshop, Premiere Pro, After Effects), Nomad Sculpt

Spoken Languages: English & Spanish (Native/Bilingual)